

POLITECNICO
MILANO 1863

054323 - INTERNET OF THINGS

POLITECNICO DI MILANO

COMPUTER SCIENCE AND ENGINEERING

IoT Challenge 2
Part 2

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1 Exercise

1.1 Given Parameters

Parameter	Value
Transmission period	$T = 2$ minutes $= 120$ seconds
Transmit energy cost	$E_{TX} = 50$ nJ/bit
Receive energy cost	$E_{RX} = 58$ nJ/bit
Payload size	12 bytes
Topic/resource length	8 bytes

Table 1: System parameters

Protocol	Message Type	Size (bytes)
CoAP	CON PUT Request	70 B
CoAP	PUT ACK	15 B
MQTT	Connect / Disconnect	60 B
MQTT	Connect ACK	50 B
MQTT	Subscribe	65 B
MQTT	Sub ACK	55 B
MQTT	Publish	75 B
MQTT	Pub ACK	50 B

Table 2: Message sizes (header + payload already included)

1.2 Energy Calculation Formula

The energy consumed to transmit or receive a message of size S (in bytes) is:

$$E = S \times 8 \times E_{TX/RX} \tag{1}$$

where the factor 8 converts bytes to bits. Specifically:

$$E_{TX}(S) = S \times 8 \times 50 \text{ nJ/bit} \tag{2}$$

$$E_{RX}(S) = S \times 8 \times 58 \text{ nJ/bit} \tag{3}$$

The total energy consumed by a device in one cycle is the sum of the energy spent transmitting and receiving all messages in that cycle:

$$E_{device}^{cycle} = \sum_{i} E_{TX}(S_i^{TX}) + \sum_{j} E_{RX}(S_j^{RX}) \tag{4}$$

1.3 EQ1a

1.3.1 Assumptions

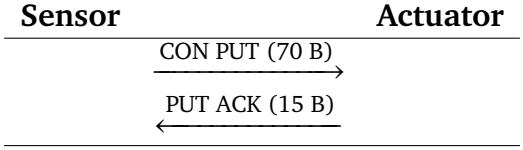
The following assumptions are made for EQ1a:

- The sensor transmits its first measurement at $t = 2$ min, for a total of **30 transmissions** in one hour.
- Since CoAP is **connectionless**, devices do not need to connect or disconnect. No setup overhead is required.
- Devices remain active throughout the entire hour and do not go to sleep between transmissions.
- **CON** messages are used, with QoS equivalent (reliable delivery via ACK).
- The channel is ideal (no losses, no retransmissions).

1.3.2 Protocol Flow

With CoAP using **Confirmable (CON)** messages, each measurement cycle consists of two messages:

- The **sensor** sends a CON PUT Request to the actuator.
- The **actuator** responds with a PUT ACK.



1.3.3 Messages per Cycle

Message	Direction	Size	Device action
CON PUT Request	Sensor → Actuator	70 B	Sensor TX, Actuator RX
PUT ACK	Actuator → Sensor	15 B	Actuator TX, Sensor RX

Table 3: CoAP messages per cycle

1.3.4 Energy per Cycle

Sensor (transmits CON PUT, receives PUT ACK):

$$E_{sensor}^{cycle} = E_{TX}(70) + E_{RX}(15) \tag{5}$$

$$E_{sensor}^{cycle} = (70 \times 8 \times 50) + (15 \times 8 \times 58) = 28000 + 6960 = 34960 \text{ nJ} \tag{6}$$

Actuator (receives CON PUT, transmits PUT ACK):

$$E_{actuator}^{cycle} = E_{RX}(70) + E_{TX}(15) \tag{7}$$

$$E_{actuator}^{cycle} = (70 \times 8 \times 58) + (15 \times 8 \times 50) = 32480 + 6000 = 38480 \text{ nJ} \tag{8}$$

Total energy per cycle:

$$E_{cycle}^{CoAP} = E_{sensor}^{cycle} + E_{actuator}^{cycle} = 34960 + 38480 = 73440 \text{ nJ} \tag{9}$$

1.3.5 Number of Cycles per Hour

The sensor transmits every 2 minutes with the first transmission at $t = 2$ min, so in 1 hour:

$$N_{cycles} = \frac{60 \text{ min}}{2 \text{ min}} = 30 \text{ cycles/hour} \tag{10}$$

1.3.6 Total Energy over 1 Hour

$$E_{1h}^{CoAP} = N_{cycles} \times E_{cycle}^{CoAP} = 30 \times 73440 = 2203200 \text{ nJ} \tag{11}$$

Result EQ1a

$$\boxed{E_{1h}^{CoAP} = 2203.2 \ \mu\text{J}} \tag{12}$$

1.4 EQ1b

1.4.1 Assumptions

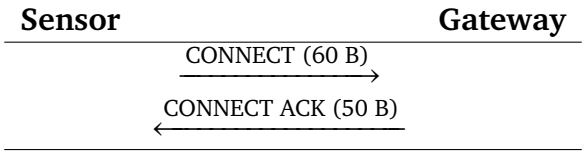
The following assumptions are made for EQ1b:

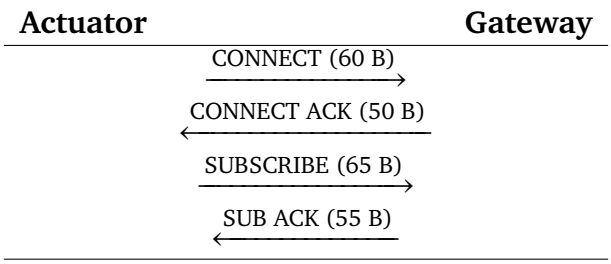
- The sensor transmits its first measurement at $t = 2$ min, for a total of **30 transmissions** in one hour.

- Devices **connect once** at the beginning of the hour and **remain connected** throughout the entire observation period without disconnecting between transmissions.
- Gateway energy consumption is **ignored** (mains-powered).
- **QoS 1** is used for both sensor and actuator.
- The actuator subscribes **once** at the beginning and remains subscribed for the entire hour.

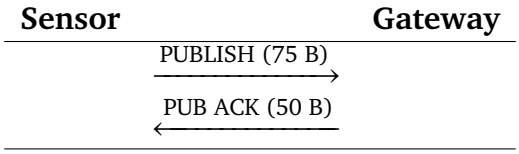
1.4.2 Protocol Flow

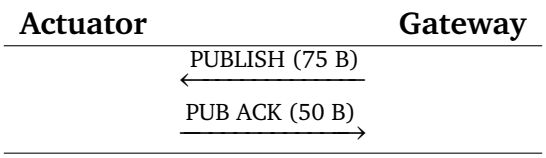
Since devices remain connected throughout the hour, the CONNECT, CONNECT ACK and SUBSCRIBE, SUB ACK messages are exchanged only **once at setup**. Each subsequent measurement cycle consists only of PUBLISH and PUB ACK messages.
Setup phase (executed once at the beginning):





Per-cycle phase (repeated 30 times):





1.4.3 Setup Energy (once)

Sensor setup:

$$E_{sensor}^{setup} = E_{TX}(60)+E_{RX}(50) = (60\times8\times50)+(50\times8\times58) = 24000+23200 = 47200 \text{ nJ} \quad (13)$$

Actuator setup:

$$E_{actuator}^{setup} = E_{TX}(60+65)+E_{RX}(50+55) = E_{TX}(125)+E_{RX}(105) \quad (14)$$

$$E_{actuator}^{setup} = (125\times8\times50)+(105\times8\times58) = 50000+48720 = 98720 \text{ nJ} \quad (15)$$

1.4.4 Energy per Cycle

Sensor (transmits PUBLISH, receives PUB ACK):

$$E_{sensor}^{cycle} = E_{TX}(75)+E_{RX}(50) = (75\times8\times50)+(50\times8\times58) = 30000+23200 = 53200 \text{ nJ} \quad (16)$$

Actuator (receives PUBLISH, transmits PUB ACK):

$$E_{actuator}^{cycle} = E_{RX}(75)+E_{TX}(50) = (75\times8\times58)+(50\times8\times50) = 34800+20000 = 54800 \text{ nJ} \quad (17)$$

Total energy per cycle:

$$E_{cycle}^{MQTT} = E_{sensor}^{cycle}+E_{actuator}^{cycle} = 53200+54800 = 108000 \text{ nJ} \quad (18)$$

1.4.5 Number of Cycles per Hour

$$N_{cycles} = \frac{60 \text{ min}}{2 \text{ min}} = 30 \text{ cycles/hour} \quad (19)$$

1.4.6 Total Energy over 1 Hour

$$E_{1h}^{MQTT} = E_{sensor}^{setup}+E_{actuator}^{setup}+N_{cycles}\times E_{cycle}^{MQTT} \quad (20)$$

$$E_{1h}^{MQTT} = 47200+98720+30\times108000 \quad (21)$$

$$E_{1h}^{MQTT} = 145920+3240000 = 3385920 \text{ nJ} \quad (22)$$

Result EQ1b

$$\boxed{E_{1h}^{MQTT} = 3385.92\ \mu\text{J}} \quad (23)$$

1.5 EQ2

1.5.1 New Scenario

The actuator wakes up only **twice per hour**: at $t = 30$ min and $t = 60$ min. The sensor still transmits every 2 minutes (30 transmissions per hour). During the 28 minutes in which the actuator sleeps, it is unreachable.

1.5.2 Protocol choice: MQTT with retained messages

The chosen protocol is **MQTT** with the following parameters:

- **QoS = 1**: reliable delivery between sensor and broker.
- **retain = 1**: the broker persistently stores the most recent value published by the sensor.
- **clean session = 1**: the actuator subscribes at each wake-up and immediately receives the last retained value.
- The sensor remains **always connected** to the broker and publishes every 2 minutes without interruption.

1.5.3 Why Not CoAP?

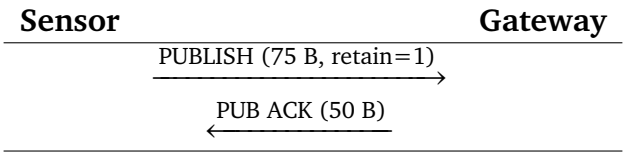
With CoAP using **CON** messages, when the actuator sleeps it cannot acknowledge the sensor's transmissions. According to the given parameters (RTO = 30 s, MAX_TRANSMISSIONS = 3), for each unacknowledged message the sensor performs 3 transmission attempts before giving up, causing severe energy waste:

- In 30 minutes, **14 messages** are sent while the actuator sleeps.
- Each unacknowledged message causes **3 retransmissions**.
- Total wasted transmissions: $14 \times 3 = 42$ unnecessary CON PUT messages.

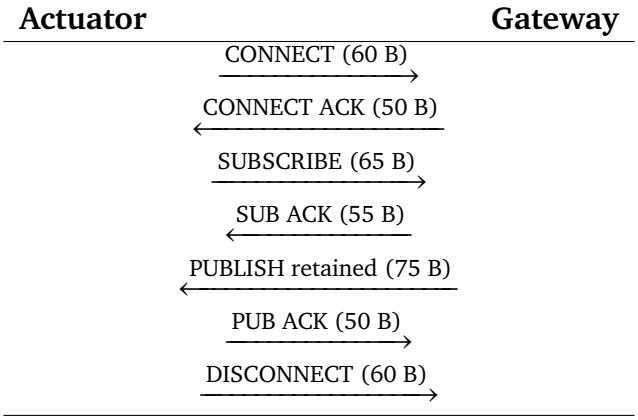
Furthermore, CoAP has no native buffering mechanism: even if the sensor retransmits successfully, data sent while the actuator sleeps is permanently lost. MQTT solves both problems: the sensor publishes without retransmission issues (the broker always acknowledges) and the broker stores the latest value via the retain flag until the actuator wakes up.

1.5.4 Protocol Flow

Sensor - always connected, publishes every 2 minutes (same as EQ1b):



Actuator - at each wake-up (CONNECT + SUBSCRIBE + receive retained + DISCONNECT):



The DISCONNECT message is **non-acknowledged**: no response is expected from the broker and the actuator immediately enters sleep mode after sending it.

1.5.5 Performance metric optimized

The chosen approach optimizes the **total energy consumption** of sensor and actuator, maximizing battery lifetime. The sensor avoids energy-intensive retransmissions, and the actuator only wakes up when needed, receiving exactly one message per wake-up.

Answer EQ2

Chosen protocol: MQTT
Parameters: QoS = 1, retain = 1, clean session = 1
MQTT with retained messages is preferred over CoAP because it eliminates retransmission overhead toward a sleeping actuator and provides native broker-side buffering. The sensor stays always connected and publishes uninterruptedly. On wake-up, the actuator connects, subscribes and instantly receives the last retained value before disconnecting.
Performance metric optimized: total energy consumption (battery lifetime).